

# IEOR 4101: Probability, Statistics and Simulation

## Lecture 6: Expectation, Variance and Covariance

# Expectations of Functions of RV

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- Often may be interested in expectation of a function of a random variable
- In general:

$$E[f(X)] = \sum_x f(x)P(X = x) = \sum_x f(x)p(x)$$

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- **Example.** R.v.  $X$  has the p.m.f  $p(2) = 0.2$ ,  $p(1) = 0.5$ ,  $p(-1) = 0.3$ . Compute  $E[X^2]$ .

$$\begin{aligned} E[X^2] &= 2^2 \times 0.2 + 1^2 \times 0.5 + (-1)^2 \times 0.3 \\ &= 1.6 \end{aligned}$$

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- **Solution.**  $E[X^2] = \sum_x x^2 P(X = x) = 2^2 P(X = 2) + 1^2 P(X = 1) + (-1)^2 P(X = -1) = 1.6$

# Expectations of Functions of RV

- Also, for any constant  $a$  and  $b$ ,

$$E[aX + b] = aE[X] + b$$

$$\begin{aligned} E[aX + b] &= \sum_x (ax + b) p(x) \\ &= a \sum_x x p(x) + b \sum_x p(x) \\ &= aE[X] + b \end{aligned}$$

- Question: is  $E[f(X)] = f(E[X])$  in general? E.g., is  $E[X^2] = (E[X])^2$ ?

No

No

# Expectations of Functions of RV

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$$E[aX + b] = aE[X] + b$$

- Question: is  $E[f(X)] = f(E[X])$  in general? E.g., is  $E[X^2] = (E[X])^2$ ?
- Try with previous example:
- $E[X^2] = 1.6$
- $E[X] = 2*0.2 + 1*0.5 + (-1)*0.3 = 0.6$
- $(E[X])^2 = 0.6^2 = 0.36$
- Answer is no

# Variance of Random Variables

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- Write  $\mu = E[X]$
- Definition: *Variance* of  $X$ , denoted by  $\text{Var}(X)$ , is defined to be
$$\text{Var}(X) = E[(X - \mu)^2]$$
- Standard deviation =  $\sqrt{\text{Var}(X)}$

# Variance of Random Variables

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$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mu)^2] \\ &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] + \mathbb{E}[-2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \end{aligned}$$

- This gives an alternate way to compute  $\text{Var}(X)$
- Note that  $\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2$  in general
- **Example.**  $\text{Var}(X)$  where  $X$  is the outcome of a fair die roll

$$X = 1, 2, \dots, 6 \quad \text{w.p.} \quad \frac{1}{6} \text{ each}$$

$$\text{i.e.} \quad p(1) = \dots = p(6) = \frac{1}{6}$$

To calculate  
 $\text{var}(X)$ ,

$$\textcircled{1} \mu = E[X] = 1 \times \frac{1}{6} + \dots + 6 \times \frac{1}{6} = 3.5$$

$$\text{var}(X) = E[(X - \mu)^2] = (1 - 3.5)^2 \times \frac{1}{6} + (2 - 3.5)^2 \times \frac{1}{6} + \dots + (6 - 3.5)^2 \times \frac{1}{6}$$

$$\textcircled{2} \quad \text{var}(X) = E[X^2] - (E[X])^2$$

$$= 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + \dots + 6^2 \times \frac{1}{6}$$

$$- (3.5)^2$$

# Properties of Variance

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$   $\leftarrow \text{Var}(aX + b) = E[(aX + b - (aE[X] + b))^2]$   
 $= a^2 E[(X - E[X])^2]$   
 $= a^2 \text{Var}(X)$
- Variance of a constant is 0  
 $\nwarrow \text{Var}(b) = E[(b - b)^2] = 0$
- What about  $\text{Var}(X + Y)$ ? Is it true that  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ ?  
No, unless  $X$  and  $Y$  are uncorrelated.

# Properties of Variance

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- $\text{Var}(aX + b) = a^2\text{Var}(X)$
- Variance of a constant is 0
- What about  $\text{Var}(X + Y)$ ? Is it true that  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ ?
- If  $X = Y$ ,  $\text{Var}(X + X) = \text{Var}(2X) = 4\text{Var}(X)$ , but  $\text{Var}(X) + \text{Var}(X) = 2\text{Var}(X)$ , so no

# Covariance

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- Definition: the *covariance* of two random variables  $X$  and  $Y$  is defined to be

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)],$$

where  $\mu_X = E[X]$  and  $\mu_Y = E[Y]$

- Covariance measures how two random variables tend to “move” together
- Useful properties:
  - $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$
  - $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
  - $\text{Cov}(X, X) = \text{Var}(X)$
  - $\text{Cov}(aX, Y) = a\text{Cov}(X, Y)$
  - $\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$
  - $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$

# Covariance

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- **Example.** Recall a previous example where the pair of  $X$  and  $Y$  has joint pmf as follows:

|                       | $Y = 2$ (savvy)  | $Y = 1$ (experienced) | $Y = 0$<br>(novice) |
|-----------------------|------------------|-----------------------|---------------------|
| $X = 0$ (online)      | $8\% \times 2/3$ | $55\% \times 2/3$     | $37\% \times 2/3$   |
| $X = 1$ (traditional) | $4\% \times 1/3$ | $29\% \times 1/3$     | $67\% \times 1/3$   |

- What is  $\text{Cov}(X, Y)$ ?
- What is  $\text{Var}(X+Y)$ ?

To calculate  $\text{Cov}(X, Y)$ ,

$$\textcircled{1} \quad P_X(0) = 8\% \times \frac{2}{3} + 55\% \times \frac{2}{3} + 37\% \times \frac{2}{3}$$
$$P_X(1) = 4\% \times \frac{1}{3} + \dots$$

$$\mu_X = E[X] = 0 \times P_X(0) + 1 \times P_X(1) = \dots$$

$$P_Y(2) = 8\% \times \frac{2}{3} + 4\% \times \frac{1}{3}$$

$$P_Y(1) = \dots$$

$$P_Y(0) = \dots$$

$$\mu_Y = E[Y] = 2 \times P_Y(2) + 1 \times P_Y(1) + 0 \times P_Y(0) = \dots$$

$$\text{Cov}(x, y) = E[(x - \mu_x)(y - \mu_y)]$$

$$= (0 - \mu_x) \times (2 - \mu_y) \times 8\% \times \frac{2}{3}$$

$$+ (0 - \mu_x) \times (1 - \mu_y) \times 55\% \times \frac{2}{3}$$

+

,

⋮

$$+ (1 - \mu_x)(0 - \mu_y) \times 67\% \times \frac{1}{3} = \dots$$

$$\textcircled{2} \quad \text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= 0 \times 2 \times 8\% \times \frac{2}{3} + 0 \times 1 \times 55\% \times \frac{2}{3}$$

$$+ \dots + 1 \times 0 \times 67\% \times \frac{1}{3}$$

$$= \mu_X \cdot \mu_Y$$

$$= \dots$$

# Covariance

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- Definition: *Correlation* between X and Y is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

- What is  $\text{Corr}(X, Y)$  in the example?

Find  $\text{var}(X)$ ,  $\text{var}(Y)$

e.g.

$$\text{var}(X) = 0^2 \times p_X(0) + 1^2 \times p_X(1) - \mu_X^2 = \dots$$

$$\text{var}(Y) = \dots$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{var}(X)\text{var}(Y)}}$$

Correlation takes values between  $-1$  and  $1$ .

• If  $\text{Corr}(X, Y) \approx 1$ , then  $X$  and  $Y$  are very positively correlated. If  $\text{Corr}(X, Y) = 1$ , then  $X$  and  $Y$  are perfectly positively correlated.

• If  $\text{Corr}(X, Y) \approx -1$ , then  $X$  and  $Y$  are very negatively correlated. If  $\text{Corr}(X, Y) = -1$ , then  $X$  and  $Y$  are perfectly negatively correlated.

If  $\text{Corr}(X, Y) = 0$  (i.e.  $\text{Cov}(X, Y) = 0$ ),

then  $X$  and  $Y$  are uncorrelated.

# Covariance

- If  $X$  and  $Y$  are independent,  $\text{Cov}(X, Y) = 0$ , and  $E[XY] = E[X]E[Y]$
- If  $X$  and  $Y$  are independent, then  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- Question:  $X$  and  $Y$  are independent. What're the expectation and variance of  $X - Y$ ?

If  $X$  and  $Y$  are uncorrelated,  
then  $\text{Cov}(X, Y) = 0$ , and so  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$  still holds.

However, if  $X$  and  $Y$  are  
uncorrelated, i.e.  $\text{Cov}(X, Y) = 0$ ,  
 $X$  and  $Y$  are not necessarily  
independent.

if  $X$  and  $Y$  are indep.,

$$E[X - Y] = E[X] - E[Y]$$

because

$$\begin{aligned} E[X - Y] &= E[X + (-1) \cdot Y] \\ &= E[X] + E[(-1) \cdot Y] \\ &= E[X] + (-1) E[Y] \\ &= E[X] - E[Y] \end{aligned}$$

$$\text{var}(X - Y) = \text{var}(X) + \text{var}(Y)$$

because

$$\text{var}(X - Y) = \text{var}(X + (-1)Y)$$

$$= \text{var}(X) + \text{var}((-1)Y) + 2\text{Cov}(X, (-1)Y)$$

$$= \text{var}(X) + (-1)^2 \text{var}(Y) + 2(-1)\text{Cov}(X, Y)$$

$$= \text{var}(X) + \text{var}(Y)$$

since  $X, Y$  are  
indep  $\Rightarrow \text{Cov}(X, Y) = 0$

if  $x$  and  $y$  are not necessarily independent,  
then

$$E(x-y) = E(x) - E(y)$$

still holds but

$$\text{Cov}(x-y) = \text{var}(x) + \text{var}(y) - 2\text{Cov}(x, y)$$